25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Advanced Functions, Polynomials & Logarithms Practice Exercises

1 MCQ

The graph of the function $f(x) = (x-1)^2 - 4$ is shown in the standard (x, y) coordinate plane below. What is the range of f?

A $[-4, \infty)$
B $[1, \infty)$
C $(-\infty, -4]$
D $(-\infty, \infty)$

2 MCQ

The polynomial function $P(x) = 2x^3 + ax^2 - 5x + b$ has $(x-1)$ as a factor. When $P(x)$ is divided by $(x+2)$, the remainder is -18. What is the remainder when $P(x)$ is divided by $(x-3)$?

A 2
B 8
C 18
D -30

3 MCQ

A function f is defined for all real numbers except $x = 0$ and $x = 1$. For all values of x in its domain, f satisfies the functional equation

$$f(x) + f\left(\frac{x-1}{x}\right) = 1 + x.$$

What is the value of $f(2)$?

A $\frac{3}{4}$
B $\frac{5}{4}$
C $\frac{7}{4}$
D $\frac{9}{4}$

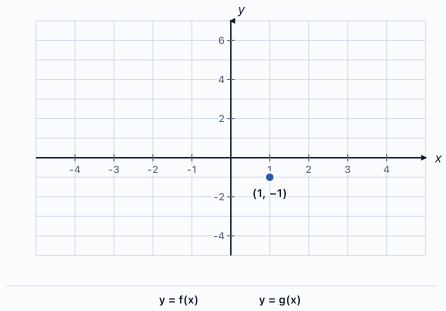
4 MCQ

Given the functions $f(x) = 3x - 4$ and $g(x) = x^2 + 5$, what is the value of $f(g(2))$?

A 23
B 9
C 18
D 27

5 MCQ

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown in the standard (x, y) coordinate plane below, with vertex $(2, -1)$ and y-intercept $(0, 3)$.
The function g is defined by $g(x) = f(x + k) - 8$, where k is a constant. If the graph of g passes through the origin $(0, 0)$, which of the following could be the value of k?



A -5

B -3

C 1

D 5

6 MCQ

What is the minimum value of the function

$$f(x) = \sqrt{x^2 - 4x + 13} + \sqrt{x^2 - 14x + 130}$$

for all real numbers x?

A 12

B $4\sqrt{10}$

C 13

D $\sqrt{170}$

7 MCQ

The function f is defined by $f(x) = \frac{3x+2}{x-4}$ for all real numbers $x \neq 4$. What is the value of $f^{-1}(2)$?

A -10

B -6

C -4

D 6

8 MCQ

The polynomial function $P(x) = x^3 + ax^2 + bx + 12$, where a and b are real constants, has $x - 1$ as a factor. When $P(x)$ is divided by $x - 2$, the remainder is -10. What is the greatest root of the equation $P(x) = 0$?

A 4

B 3

C 1

D 6

9 MCQ

A line L is tangent to the graph of the polynomial function $f(x) = x^4 - 6x^2 + 5x + 7$ at two distinct points in the xy-plane. What is the y-intercept of line L?

A 2

B -2

C 7

D 16

10 MCQ

In the standard (x, y) coordinate plane, the graph of $y = f(x)$ contains the point $(-2, 5)$. The function g is defined by $g(x) = f(x - 3) - 4$. Which of the following points must lie on the graph of $y = g(x)$?

A $(-5, 1)$

B $(1, 1)$

C $(1, 9)$

D $(-5, 9)$

11 MCQ

If $f(x) = 5x + 4$ and $g(x) = -4x$, find $f(g(2))$.

A -36

B 6

C -56

D -31

12 MCQ

If $f(x) = 2x - 7$ and $g(x) = 4x - 1$, find $f(g(-2))$.

A -19

B -45

C -23

D -25

13 MCQ

If $f(x) = -2x + 7$ and $g(x) = 4x - 3$, find $f(g(1))$.

A 5

B 9

C 17

D 3

14 MCQ

If $f(x) = -3x + 5$ and $g(x) = 3x + 5$, find $f(g(5))$.

A 5

B -55

C -25

D -58

15 MCQ

If $f(x) = -5x + 8$ and $g(x) = -x + 6$, find $f(g(-1))$.

A 14

B -7

C -32

D -27

16 MCQ

If $f(x) = 2x + 7$ and $g(x) = 3x - 2$, find $f(g(-5))$.

A -18

B -27

C -11

D -25

17 MCQ

If $f(x) = -3x - 5$ and $g(x) = -4x$, find $f(g(-1))$.

A 2

B -17

C 8

D -20

18 MCQ

If $f(x) = -5x - 3$ and $g(x) = -x - 3$, find $f(g(-4))$.

A -8

B 21

C -20

D -13

19 MCQ

If $f(x) = -2x - 2$ and $g(x) = 2x - 3$, find $f(g(-1))$.

A -2

B 8

C -3

D 6

20 MCQ

If $f(x) = 5x + 3$ and $g(x) = -x - 2$, find $f(g(-1))$.

A -1

B 0

C 3

D -2

21 MCQ

If $f(x) = -2x + 3$ and $g(x) = -4x + 3$, find $f(g(-2))$.

A -25

B -21

C -19

D 15

22 MCQ

If $f(x) = x + 4$ and $g(x) = 4x + 6$, find $f(g(4))$.

A 27

B 24

C 26

D 38

23 MCQ

If $f(x) = x - 3$ and $g(x) = -x + 2$, find $f(g(0))$.

A -3

B 5

C -1

D 0

24 MCQ

If $f(x) = -2x + 2$ and $g(x) = -4x - 5$, find $f(g(5))$.

A -28

B 27

C 52

D 50

25 MCQ

If $f(x) = 3x + 4$ and $g(x) = x - 5$, find $f(g(-1))$.

A 0

B -14

C -4

D -11